AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII EXAMINATION

APRIL/ MAY 2019

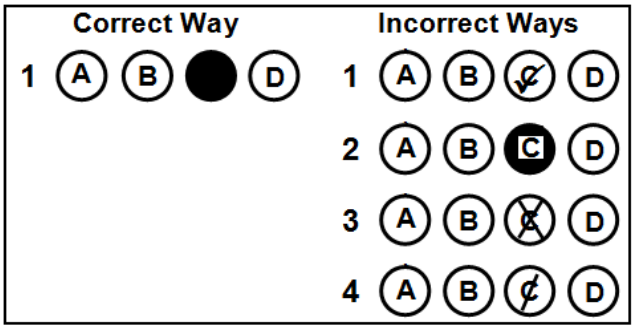
Computer Science Paper I

Time: 45 minutes Marks: 30

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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 30 only. From question number 13 to 30, there are two sections. Attempt EITHER section I (C Language) OR section II (Visual Basic).
4. In each question, there are four choices A, B, C, D. Choose ONE. On the answer grid, black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.

1. Which of the following is TRUE about records in a database?

   A. A record can be set as primary key of a table.
   B. The sort option cannot be applied on a record.
   C. The filter option cannot be applied on a record.
   D. A record stores one piece of data about an item.

2. A query fetches the information from the database when

   I. there is numeric data to be fetched
   II. all criteria used in the query to fetch data are true
   III. there are multiple tables from which data is to be fetched

   A. I only.
   B. II only.
   C. I and III.
   D. II and III.

3. An example of valid field name in MS Access is

   A. Total[Earning
   B. Total!Earning
   C. Total.Earning
   D. Total Earning

4. Naveed has applied a database relationship on two tables **M** and **N**. In this relationship, one row of table **M** is linked with multiple rows of table **N**, but in table **N**, one row can only be linked to one row of table **M**.

   The relationship that Naveed has applied is

   A. one-to-one.
   B. one-to-many.
   C. many-to-one.
   D. many-to-many.

5. A relationship is created between two tables, i.e. student and fee. The student_id field from the student table is used to build this relationship.

   According to the given description, student_id field is

   A. foreign key in both students and fee tables.
   B. primary key in both students and fee tables.
   C. foreign key in students table and primary key in fee table.
   D. primary key in students table and foreign key in fee table.

6. The database view that allows to set the data types of each field and select a primary key for the table is

   A. design view.
   B. datasheet view.
   C. pivot table view.
   D. pivot chart view.

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7. Which of the following characteristics is FALSE about primary key?

A. Its value changes rarely.
B. It is never null or empty.
C. It uniquely identifies each record in a table.
D. It is applied to a field having any data type.

8. Consider the given data.

3^2 5^2 1^5 2^3 4^3

If descending sort is applied on this data, then the result is

1^5 2^3 3^2 5^2 4^3	4^3 5^2 3^2 2^3 1^5
A	B
5^2 4^3 3^2 2^3 1^5	1^5 2^3 3^2 4^3 5^2
C	D

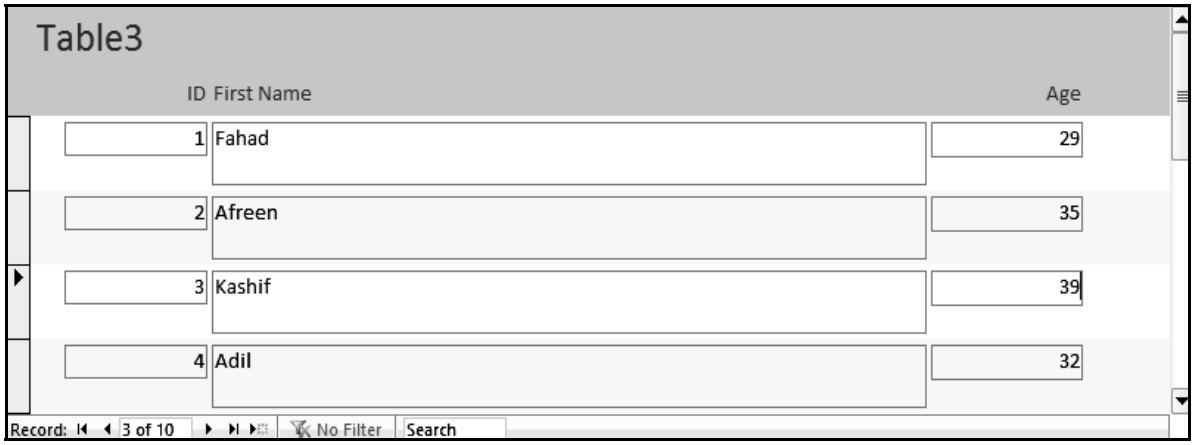
9. Omar finalised the structure of a database table in design view. He has not entered any data yet in the table.

The first row of this table in datasheet view will be marked with

A. *
B. $
C. **
D. $$

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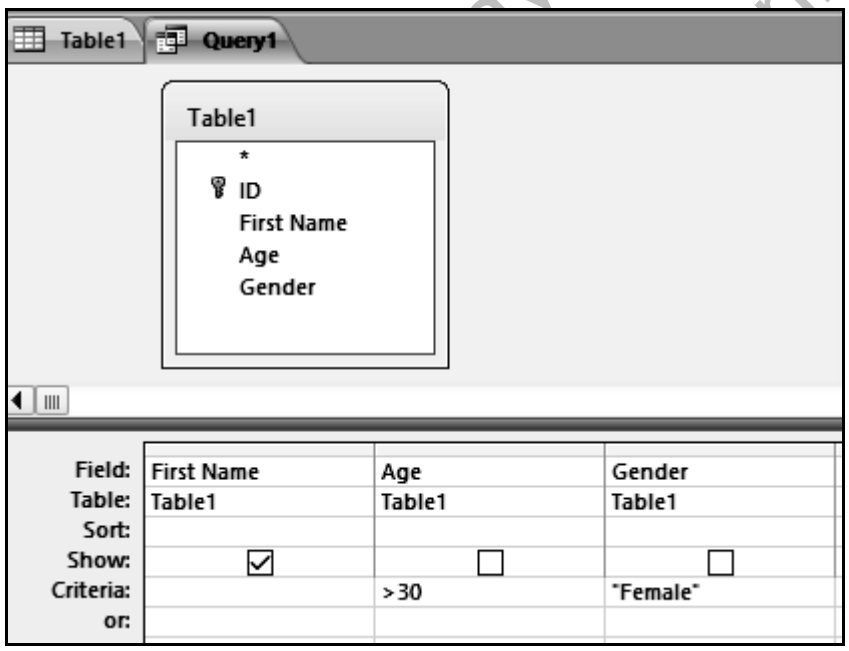
10.



The given form is an example of

A. tabular form.
B. justified form.
C. datasheet form.
D. columnar form.

11. When the given query is run, then the output would be shown in

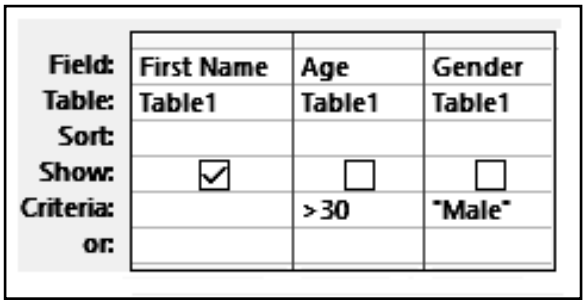


A. SQL view.
B. design view.
C. datasheet view.
D. pivot table view.

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12. Consider the given table and query applied on it in database.

ID	First Name	Age	Gender
1	Atif	29	Male
2	Ammara	35	Female
3	Omar	39	Male
4	Sajid	24	Male
5	Rashid	52	Male



The output of this query would be

A

First Name	Age	Gender
Omar	39	Male
Rashid	52	Male

B

First Name
Omar
Rashid

C

First Name	Age	Gender
Ammara	35	Female
Omar	39	Male
Rashid	52	Male

D

First Name
Ammara
Omar
Rashid

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ATTEMPT EITHER SECTION I OR II

Section I: C Language

13. Which of the following is FALSE about assembly language?

A. It takes less memory space as compared to high level languages.
B. Debugging of programs is simplified as compared to machine code.
C. Before running on a machine, assembly language program requires a translator.
D. Each assembly language instruction generates many machine language instructions.

14. One of the features of C language is that it

A. is a highly structured language.
B. is difficult for humans to understand.
C. uses an assembler for translation purposes.
D. is a type of low level programming language.

15. Each statement in a C program is terminated by using

A. a semicolon.
B. a hash symbol.
C. a forward slash.
D. the inverted commas.

16. The given code contains few errors.

```
#include <stdio.h>
int main()
{
 integer a = 0;
 return 0;
}
```

The number of errors in this code is

A. 2
B. 3
C. 4
D. 5

17. The int before main function in **int main(void)** indicates that the main function will

A. return a float value.
B. need a float argument.
C. return an integer value.
D. need an integer argument.

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18. The data type that can store the largest numeric value with decimal point is

A. int.
B. long.
C. float.
D. double.

19. The output of the given C language code will be

```
int    a = 25;
int    b = 2;
printf ("Result  =  %d" , 6 + a / 5 * b)
```

A. Result = 16
B. Result = 8.5
C. Result = 6 + a / 5 * b
D. Result = %6 + a / 5 * b

20. The number of repetitions due to the given **for loop** statement are

for (a = 0; a<=13; a = a + 2)

A. 6
B. 7
C. 13
D. 14

21. Read the given incomplete C program.

```
#include <stdio.h>
void main(void)
{
        int x = 3, y = 3, z = 3;
        x = 2 * y + z;
        y = x - (2 * z) / 3;
        z = x / y;
```

The remaining part of this code to give the output 1 is

printf("Result = %d",(x < y) || (x < z)); }	printf("Result = %d",(x < y) || (x > z)); }
A	B
printf("Result = %d",(x < y) || (y < z)); }	printf("Result = %d",!((x < y) || (y > z))); }
C	D

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22. The given C program will input an integer value, calculate its cube and output this value. However, there are few errors in this program.

(**Note**: The cube of integer **a** will be represented as **a × a × a**)

```
#include <stdio.h>
void main(void)
{
        int x=0;
        scanf("%.1f", x);
        x = x * x ;
        printf("Result = %.1f", x);
}
```

The correct version of the given program would be

`#include <stdio.h>`
`void main(void)`
`{`
`  int x=0;`
`  scanf("%.1f", &x);`
`  x = x * x * x;`
`  printf("Result = %.1f", x);`
`}`
	`#include <stdio.h>`
	`void main(void)`
	`{`
	`  int x=0;`
	`  scanf("%d", x);`
	`  x = x * x * x;`
	`  printf("Result = %d", x);`
	`}`

A	B

`#include <stdio.h>`
`void main(void)`
`{`
`  int x=0;`
`  scanf("%.1f", x);`
`  x = x * x * x;`
`  printf("Result = %.1f", x);`
`}`
	`#include <stdio.h>`
	`void main(void)`
	`{`
	`  int x = 0;`
	`  scanf("%d", &x);`
	`  x = x * x * x;`
	`  printf("Result = %d", x);`
	`}`

C	D

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23. Read the given C program.

```
#include <stdio.h>
int main(void)
{
printf("Hello\nWorld");
return 0;
}
```

The output of the given C program will be

"Hello\nWorld"	Hello\nWorld	Hello World	Hello /nWorld
A	B	C	D

24. Read the given C program.

```
#include <stdio.h>
void main(void)
{
   int a, b, c = 0;
     for(b =1; b <= 5; b++)
          {
  scanf("%d", &a);
  if( (a + 5) % 3 == 0)
  c = c + a;
          }
   printf("result = %d", c);
}
```

If the input of this program is the sequence of five numbers (1, 2, 3, 4, 5), then the output will be

A. result = 0
B. result = 5
C. result = 10
D. result = 15

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25. Read the given C program.

```
#include <stdio.h>
void main( )
{
  int x = 3;
  if(x++ == 3 && x++ > 3)
    printf("%d", x);
  else
    printf("%d", x);
}
```

The output of the given program is

A. 3
B. 4
C. 5
D. 6

26. Read the given C program.

```
#include <stdio.h>
int main( )
{
        int a = 10, b = 15;
        while(a < ++b)
        {
        printf("\nPakistan");
        a = a + 2;
        }
        return 0;
}
```

The number of times 'Pakistan' would be printed on execution of this program is

A. five.
B. six.
C. ten.
D. fifteen.

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Refer the given C program to answer Q.27 and Q.28.

```
#include <stdio.h>
int Z(int x);
void main( )
{
   int b = Z(6);
   printf("%d", b);
}

int Z(int x)
{
   int c;
         c = (x * 5) % 7;
   return c;
}
```

27. In the given program, the code line(s) that define(s) the function Z( ) is/ are

    I. int b = Z(6);
    II. int Z(int x);
    III. int Z(int x)

    A. I only.
    B. III only.
    C. I and II.
    D. II and III.

28. The output of the given program would be

    A. 2
    B. 6
    C. 30
    D. 35

29. Which of the following is FALSE about functions in C programming?

    A. They allow dividing a complex program into a number of tasks.
    B. They increase the reusability of the codes.
    C. They make debugging of a program easy.
    D. They depend on each other.

30. The C language command that opens a text file named newfile.txt in the read mode is

    A. fopen ("r", "newfile.txt");
    B. fopen ("newfile.txt" , "r");
    C. fopen ("newfile.txt" , "read");
    D. fopen ("read", "newfile.txt");

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ATTEMPT EITHER SECTION I OR II

Section II: Visual Basic

13. In Visual Basic, the default toolbar that appears below the menu bar is known as

A. edit toolbar.
B. debug toolbar.
C. form editing toolbar.
D. standard toolbar.

14. The keyword used to comment a code line in Visual Basic is

A. Rem.
B. Com.
C. Remark.
D. Comment.

15. A file extensions that is used for ActiveX control files is

A. .OCX
B. .EXE
C. .FRM
D. .BAS

16. Which of the following supports the activated event?

A. Form
B. Label
C. Button
D. Text Box

17. Read the given code.

```
Dim x As Integer = 20
Dim y As Double = 3
Dim z As Double
z = x Mod y
```

The result of this code will be

A. 17
B. 6
C. 2
D. 23

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18. While considering the purpose of data types in Visual Basic, the appropriate data type to store the address of a person is

A. byte.
B. long.
C. single.
D. string.

19. The Visual Basic statement that outputs 1 is

A. Print 1 XOR 0
B. Print 1 And 0
C. Print Not 1 And Not 0
D. Print Not 1 XOR Not 0

20. The number of byte(s) occupied by the integer data type is

A. 1
B. 2
C. 4
D. 8

21. The CORRECT Visual Basic code that would output **ram** is

Dim a, b As String a = "Program" b = (a, 3) Right Print b	Dim a, b As String a = "Program" b = Right (a, 3) Print b
A	B
Dim a, b As String a = "Program" b = Left (a, 4) Print b	Dim a, b As String a = "Program" b = (a, 4) Left Print b
C	D

22. The CORRECT Visual Basic code line to call a user-defined function named as AddNumber() is

A. Public Function AddNumber(n1, n2 As Integer) As Integer
B. Public Function AddNumber(n1, n2 As Integer)
C. AddNumber(n1, n2 As Integer)
D. AddNumber(n1, n2)

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23. The Visual Basic code that will print number series 3, 7, 11, 15 on screen is

For count As Integer = 1 To 15 Step +1 count = count + 3 Print count Next	For count As Integer = 1 To 15 Step +1 count = count + 2 Print count Next
A	B
For count As Integer = 1 To 15 Step +2 count = count + 2 Print count Next	For count As Integer = 1 To 15 Step +2 count = count + 1 Print count Next
C	D

24. Read the given incomplete Visual Basic code.

```
Else
Console.WriteLine(X)
End If
Next
```

The first three lines of code that should come before Else in the above given code to print integers 1, 3, 5, 7 on screen is

For X As Integer = 1 to 10 Step +2 If X < 7 then Exit For	For X As Integer = 1 to 10 Step +2 If X > 7 then Exit For
A	B
For X As Integer = 1 to 10 Step +2 Exit For If X > 7 then	For X As Integer = 1 to 10 Step +2 Exit For If X < 7 then
C	D

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25. Read the given Visual Basic code.

```
Dim G As Integer = 10
Do While G >= 0
G = G - 3
Loop
```

After the loop ends, the value of **G** will be

A. 0
B. 3
C. -2
D. -3

26. The given code will display the grades in a text box.

```
If Average  > 80 Then
txtGrade.Text = "A"
ElseIf Avg > 70 Then
txtGrade.Text = "B"
ElseIf Avg > 60 Then
txtGrade.Text = "C"
ElseIf Avg > 50 Then
txtGrade.Text = "D"
Else
txtGrade.Text = "E"
End If
```

If the Average is 50, then the grade this Visual Basic code would display is

A. D
B. E
C. CD
D. AB

27. As compared to functions of other objects of ADO.NET, the main function of data-reader object is to

A. provide the physical connection to the data source.
B. retrieve query results from the data source which do not update the data.
C. populate a dataset object with the results retrieved from a data source.
D. use stored procedures that execute at the data source.

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28. In traditional data binding, the text property of the text box on a form is bound to a

A. column of a database.
B. table of a database.
C. row of a database.
D. relationship of a database.

29. The CORRECT code line to jump to the first record in data source named as dtaBooks is

A. MoveFirst.dtaBooks.DataSource
B. MoveFirst.dtaBooks.Recordset
C. dtaBooks.DataSrouce.MoveFirst
D. dtaBooks.Recordset.MoveFirst

30. The CORRECT command to display a yellow coloured dot at the coordinates x = 500 and y = 250 is

A. Pset (500, 250, Yellow)
B. Pset (500, 250), VbYellow
C. Pset (500, 250, VbYellow)
D. Pset (500, 250), Yellow

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END OF PAPER

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